## 0.1 Formal Definition of the Cybernetic Automaton

This section presents a formal definition of the new model, cybernetic automata. A more general and higher-level discussion of the new model can be found in the next section.

**DEFINITION 1** A Cybernetic Automaton, $A$, is defined by

$$A \equiv \langle Q, \Sigma, \Delta, \delta, \lambda, q_0, R, P, C, E \rangle$$

where $Q$ is the set of states, $\Sigma$ is the input alphabet, $\Delta$ is the output alphabet, $\delta : Q \cup \{\epsilon\} \times \Sigma \mapsto Q$ is the state transition function, $\lambda : Q \times \Sigma \mapsto \Delta$ where $\lambda(q, a) = \sigma\left(P^{\Delta}_{q,a}\right)$ is the output selection function, $q_0$ is the initial state, $R \subseteq Q$ is the set of reward states, $P \subseteq Q$ is the set of punishment states, $C : Q \times \Sigma \mapsto \Re$ is a confidence function over the set of transitions, and $E : Q \times \Sigma \times Q \times \Sigma \mapsto \Re$ is an expectation function between pairs of transitions.

The operation of the cybernetic automaton is described by the following procedures.

Cybernetic Automaton Operation:

1. Let $c = q_0$, $q_l = q_0$, $a_l = \epsilon$, $o = \epsilon$ and $o_l = \epsilon$.

2. If the time since the last input symbol arrived is greater than $\tau$ then:

   If $\delta(c, \epsilon)$ is defined:

      If $\delta(c, \epsilon)$ is marked as temporary:

         Mark $\delta(c, \epsilon)$ as permanent.

      Let $q_l = c$.

      Let $c = \delta(c, \epsilon)$.

   Let the anchor state $q_a = c$.

   Let $a_l = \epsilon$ and $o_l = \epsilon$.

   Unmark all symbols $b$ and distributions $P^{\Delta}_{q,a}$.

   Go to step 2.

3. Input a set of symbol-strength pairs, $I = \{\langle a_1, s_1\rangle, \langle a_2, s_2\rangle, \ldots, \langle a_n, s_n\rangle\}$ and let $I_l$ be the previous set $I$.

4. Let $\langle a_d, s_d\rangle$ be the dominant input symbol-strength pair, where $s_d$ is a maximum over all $s_i$, $1 \le i \le n$.

5. Perform Create New Transitions.

6. Let $o_l = o$.

7. Output the symbol $o = \sigma\left(P^{\Delta}_{c,a_d}\right)$ with strength $\frac{s_d C(c,a_d)}{1+C(c,a_d)}$.

8. Mark $P^{\Delta}_{c,a_d}$ and its element for $o$ for later modification.

9. Perform Update Expectations.

10. Let $q_l = c$.
11. Let $c = \delta(c, a_d)$.
12. Let $a_l = a_d$.
13. If $c \in R$, then perform Apply Reward else if $c \in P$, then perform Apply Punishment else perform Apply Conditioning.
14. Go to step 2.

Create New Transitions:

If $\delta(c, \epsilon)$ is defined and marked temporary,

Remove the transition $\delta(c, \epsilon)$.

For each input pair $\langle a_i, s_i \rangle$ in $I$:

If $\delta(c, a_i)$ does not exist, then

Add a unique state $q_N$ to $Q$.
Let $\delta(c, a_i) = q_N$.
Let $\delta(q_N, \epsilon) = q_a$ and mark the transition as temporary.
If there exists a state $q'$ for which $\delta(q', a_i)$ exists, then
Let $P^{\Delta}_{c,a_i} = P^{\Delta}_{q',a_i}$.
Let $C(c, a_i) = C(q', a_i)$.
If $\delta(q', a_i) \in R$, add $q_N$ to $R$, else if $\delta(q', a_i) \in P$, add $q_N$ to $P$.
else
Let $P^{\Delta}_{c,a_i}(\epsilon) = \eta$ and $P^{\Delta}_{c,a_i}(b) = \frac{(1-\eta)}{|\Delta|-1}, b \neq \epsilon$.
Let $C(c, a_i) = 0.1$.

Update Expectations:

If $E(q_l, a_l, c, a_d)$ exists, then

Let $\Delta E(q_l, a_l, c, a_d) = \alpha(1 - E(q_l, a_l, c, a_d))$.
Let $C(q_l, a_l) = C(q_l, a_l)(1 - \beta|\Delta E(q_l, a_l, c, a_d)|)$.
Let $\Delta E(c, a_d, q_l, a_l) = \alpha(1 - E(c, a_d, q_l, a_l))$.
Let $C(c, a_d) = C(c, a_d)(1 - \beta|\Delta E(c, a_d, q_l, a_l)|)$.

else

Create and let $E(q_l, a_l, c, a_d) = \alpha$.
Create and let $E(c, a_d, q_l, a_l) = \alpha$.

For each symbol $a \in \Sigma$:

If $E(q_l, a_l, c, a)$ exists and $\langle a, \cdot\rangle \notin I$, then

Let $\Delta E(q_l, a_l, c, a) = -\alpha E(q_l, a_l, c, a)$.
Let $C(q_l, a_l) = C(q_l, a_l)(1 - \beta|\Delta E(q_l, a_l, c, a)|)$.
Let $\Delta E(c, a, q_l, a_l) = -\alpha E(c, a, q_l, a_l)$.
Let $C(c, a) = C(c, a)(1 - \beta|\Delta E(c, a, q_l, a_l)|)$.

For each $q \in Q$, $a \in \Sigma$, $q \neq q_l$ or $a \neq a_l$:

If $E(c, a_d, q, a)$ exists, then

Let $\Delta E(c, a_d, q, a) = -\alpha E(c, a_d, q, a)$.
Let $C(c, a_d) = C(c, a_d)(1 - \beta|\Delta E(c, a_d, q, a)|)$.
Let $\Delta E(q, a, c, a_d) = -\alpha E(q, a, c, a_d)$.
Let $C(q, a) = C(q, a)(1 - \beta|\Delta E(q, a, c, a_d)|)$.

For each $a \in \Sigma$ and $b \in \Sigma, a \neq b$:

If $\langle a, \cdot\rangle \in I$ and $\langle b, \cdot\rangle \in I$, then

If $E(c, a, c, b)$ exists, then
Let $\Delta E(c, a, c, b) = \alpha(1 - E(c, a, c, b))$.
Let $C(c, a) = C(c, a)(1 - \beta|\Delta E(c, a, c, b)|)$.
else
Create and let $E(c, a, c, b) = \alpha$.

else if $\langle a, \cdot\rangle \in I$ or $\langle b, \cdot\rangle \in I$, then

If $E(c, a, c, b)$ exists, then
Let $\Delta E(c, a, c, b) = -\alpha E(c, a, c, b)$.
Let $C(c, a) = C(c, a)(1 - \beta|\Delta E(c, a, c, b)|)$.

Apply Reward:

Let $t = 1$.

For each $P^{\Delta}_{(q,a)}$ which is marked, starting at the most recent and moving toward the oldest:

For each marked element, $B$, in the distribution $P^{\Delta}_{(q,a)}$:

Let $P^{\Delta}_{q,a}(B) = \frac{P^{\Delta}_{q,a}(B)+\zeta t s_d(1/C(q,a))}{1+\zeta t s_d(1/C(q,a))}$.
Let $P^{\Delta}_{q,a}(b) = \frac{P^{\Delta}_{q,a}(b)}{1+\zeta t s_d(1/C(q,a))}$ for all $b \in \Delta, b \neq B$.
Let $\Delta C(q, a) = \zeta t s_d$.
Unmark $B$ from pending update.
For each state, $q'$:
Let $P^{\Delta}_{q',a}(B) = \frac{P^{\Delta}_{q',a}(B)+\nu\zeta t s_d\left(1/C(q',a)\right)}{1+\nu\zeta t s_d(1/C(q',a))}$.
Let $P^{\Delta}_{q',a}(b) = \frac{P^{\Delta}_{q',a}(b)}{1+\nu\zeta t s_d(1/C(q',a))}$ for all $b \in \Delta, b \neq B$.

Let $\Delta C(q', a) = \nu\zeta t s_d$.

Unmark $P^{\Delta}_{(q,a)}$ from pending update.

Let $t = \kappa t$.

Apply Punishment:

Let $t = 1$.

For each $P^{\Delta}_{(q,a)}$ which is marked, starting at the most recent and moving toward the oldest:

For each marked element, $B$, in the distribution $P^{\Delta}_{(q,a)}$:

Let $P^{\Delta}_{q,a}(B) = \frac{P^{\Delta}_{q,a}(B)}{1+\zeta t s_d(1/C(q,a))}$.

Let $P^{\Delta}_{q,a}(b) = \frac{P^{\Delta}_{q,a}(b)+\left(\frac{1}{|\Delta|-1}\zeta t s_d(1/C(q,a))\right)}{1+\zeta t s_d(1/C(q,a))}$ for all $b \in \Delta, b \neq B$.

Let $\Delta C(q, a) = \zeta t s_d$.

Unmark $B$ from pending update.

For each state, $q'$:

Let $P^{\Delta}_{q',a}(B) = \frac{P^{\Delta}_{q',a}(B)}{1+\nu\zeta t s_d(1/C(q',a))}$.

Let $P^{\Delta}_{q',a}(b) = \frac{P^{\Delta}_{q',a}(b)+\left(\frac{1}{|\Delta|-1}\nu\zeta t s_d(1/C(q,a))\right)}{1+\nu\zeta t s_d(1/C(q',a))}$ for all $b \in \Delta, b \neq B$.

Let $\Delta C(q', a) = \nu\zeta t s_d$.

Unmark $P^{\Delta}_{(q,a)}$ from pending update.

Let $t = \kappa t$.

Apply Conditioning:

If $o_l \neq \epsilon$ and $o \neq o_l$, then

For each $a \in \Sigma$:

If $E(q_l, a_l, q_l, a)$ exists and $\langle a, \cdot\rangle \in I_l$, then

Let $P^{\Delta}_{q_l,a}(o_l) = \frac{P^{\Delta}_{q_l,a}(o_l)+\gamma s_d(1/C(q_l,a))}{1+\gamma s_d(1/C(q_l,a))}$.

Let $P^{\Delta}_{q_l,a}(b) = \frac{P^{\Delta}_{q_l,a}(b)}{1+\gamma s_d(1/C(q_l,a))}$ for all $b \in \Delta, b \neq o_l$.

For each $q \in Q$ and $a \in \Sigma$ such that $\delta(q, a) = q_l$:

If $E(q_l, a_l, q, a)$ exists, then

Let $P^{\Delta}_{q,a}(o_l) = \frac{P^{\Delta}_{q,a}(o_l)+\gamma s_d(1/C(q,a))}{1+\gamma s_d(1/C(q,a))}$.

Let $P^{\Delta}_{q,a}(b) = \frac{P^{\Delta}_{q,a}(b)}{1+\gamma s_d(1/C(q,a))}$ for all $b \in \Delta, b \neq o_l$.

For each $a \in \Sigma$:

If $E(q_l, a_l, q_l, a)$ exists, $\langle a, \cdot\rangle \in I_l$ and $P^{\Delta}_{q_l,a}$ has not been conditioned, then

Let $\Delta C(q_l, a) = \gamma s_d$.
Perform Update Conditioning $(q_l, a, s_d(1/C(q_l, a)))$.

For each $q \in Q$ and $a \in \Sigma$ such that $\delta(q, a) = q_l$:

If $E(q_l, a_l, q, a)$ exists and $P^{\Delta}_{q,a}$ has not been conditioned, then
Let $\Delta C(q, a) = \gamma s_d$.
Perform Update Conditioning $(q, a, s_d(1/C(q, a)))$.

Update Conditioning: $(q', a', s)$

If $s > 0$, then

For each $a \in \Sigma$:

If $E(q', a', q', a)$ exists and $P^{\Delta}_{q',a}$ has not been conditioned, then
Let $P^{\Delta}_{q',a}(o_l) = \frac{P^{\Delta}_{q',a}(o_l)+\gamma s(1/C(q',a))}{1+\gamma s(1/C(q',a))}$.
Let $P^{\Delta}_{q',a}(b) = \frac{P^{\Delta}_{q',a}(b)}{1+\gamma s(1/C(q',a))}$ for all $b \in \Delta, b \neq o_l$.

For each $q \in Q$ and $a \in \Sigma$ such that $\delta(q, a) = q'$:

If $E(q', a', q, a)$ exists, then
Let $P^{\Delta}_{q,a}(o_l) = \frac{P^{\Delta}_{q,a}(o_l)+\gamma s(1/C(q,a))}{1+\gamma s(1/C(q,a))}$.
Let $P^{\Delta}_{q,a}(b) = \frac{P^{\Delta}_{q,a}(b)}{1+\gamma s(1/C(q,a))}$ for all $b \in \Delta, b \neq o_l$.

For each $a \in \Sigma$:

If $E(q', a', q', a)$ exists, $\langle a, \cdot\rangle \in I_l$ and $P^{\Delta}_{q',a}$ has not been conditioned, then
Let $\Delta C(q', a) = \gamma s$.
Perform Update Conditioning $(q', a, s(1/C(q', a)))$.

For each $q \in Q$ and $a \in \Sigma$ such that $\delta(q, a) = q_l$:

If $E(q', a', q, a)$ exists and $P^{\Delta}_{q,a}$ has not been conditioned, then
Let $\Delta C(q, a) = \gamma s$.
Perform Update Conditioning $(q, a, s(1/C(q, a)))$.

where $\alpha$, $\beta$, $\gamma$, $\zeta$, $\nu$, $\kappa$ and $\eta$ are constants less than unity. The parameter $\alpha$ controls how fast expectations are acquired and lost, $\beta$ control how fast confidence is lost and $\gamma$ controls how fast learning takes place during conditioning. For supervised learning, the $\zeta$ parameter controls the speed of learning, $\nu$ determines how much the probability changes are propagated to other transitions with the same input symbol and $\kappa$ ensures that more recent action get more reinforcement than earlier ones. The parameter $\eta$ is the amount of probability that is assigned to the $\epsilon$ responses for a newly created transition. It must be less than unity in order for the Cybernetic Automaton Learning Theorem, described in the next chapter, to be valid. All of these parameters will be discussed in more detail in the next section. □